

ELECTRICAL & ELECTRONICS ENGINEERING

CODE	COURSE NAME	CATEGORY	L	T	P	CREDITS
EET306	POWER ELECTRONICS	PCC	3	1	0	4

Preamble: To impart knowledge about the power semiconductor devices, the operation of various power converters and its applications.

Prerequisite: Basics of Electrical Engineering / Introduction to Electrical Engineering/
Basics of Electronics Engineering

Course Outcomes: After the completion of the course the student will be able to:

CO 1	Explain the operation of modern power semiconductor devices and its characteristics.
CO 2	Analyse the working of controlled rectifiers.
CO 3	Explain the working of AC voltage controllers, inverters and PWM techniques.
CO 4	Compare the performance of different dc-dc converters.
CO 5	Describe basic drive schemes for ac and dc motors.

Mapping of course outcomes with program outcomes

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO 1	3	1	-	1	-	-	-	-	-	-	-	-
CO 2	3	2	1	2	-	-	-	-	-	-	-	2
CO 3	3	3	-	-	-	-	-	-	-	-	-	-
CO 4	3	3	2	2	-	-	-	-	-	-	-	2
CO 5	3	2	-	-	-	-	-	-	-	-	-	2

Assessment Pattern

Bloom's Category	Continuous Assessment Tests		End Semester Examination
	1	2	
Remember (K1)	10	10	20
Understand (K2)	20	20	30
Apply (K3)	20	20	50
Analyse (K4)	-	-	-
Evaluate (K5)	-	-	-
Create (K6)	-	-	-

End Semester Examination Pattern : There will be two parts; Part A and Part B. Part A contain 10 questions with 2 questions from each module, having 3 marks for each question. Students should answer all questions. Part B contains 2 questions from each module of which student should answer any one. Each question can have maximum 2 sub-divisions and carry 14 marks.

Course Level Assessment Questions

Course Outcome 1 (CO1):

1. Explain the working and switching characteristics of SCR, MOSFET, IGBT (K1)
2. Give a brief description on wide band-gap power devices (K1)
3. Draw and explain the switching characteristics of SCR (K1, K2)
4. Discuss the protection circuits for SCR (K2)
5. Explain different types of isolation in gate drive for power converter circuits (K1, K2)

Course Outcome 2 (CO2):

1. Describe the working with waveforms of single phase half wave rectifiers for different firing angles. (K1)
2. Describe the working with waveforms of single phase fully controlled rectifiers for different firing angles and loads.(K2)
3. Describe the working with waveforms of single phase half controlled rectifiers for different firing angles and loads.(K2)
4. Describe the working with waveforms of three phase rectifiers for different firing angles and loads. (K2)
5. Problems in finding the average output voltage of rectifier. (K2, K3)

Course Outcome 3 (CO3):

1. Explain the working of ACVC with R and RL loads. (K1)
2. Explain single phase inverter for R and RL loads, problems in finding the output voltage, THD of inverter. (K2, K3)
3. Explain 3 phase mode 120° and 180° conduction modes. (K4)
4. Explain single phase current source inverter PWM Inverter. (K1)
5. Explain single pulse PWM, multiple pulse, and sinusoidal PWM technique (K1, K2)

Course Outcome 4 (CO4):

1. Explain the working of step up and step down converters. (K1, K2)
2. Problems related to step up and step down converters. (K2, K3)
3. Analyse the working of Buck, Boost & Buck Boost regulators. (K3, K4)
4. Design the value of filter inductor & capacitance in regulators. (K3, K4)
5. Problems in Buck, Boost & Buck Boost regulators. (K2, K3)

Course Outcome 5 (CO5):

1. Explain the block diagram of an electric drive (K1,K2)
2. Explain the working of single phase rectifier fed DC drive (K2, K3)
3. Explain the chopper controller DC drive (K2,K3)
4. Explain the four quadrant operation of a DC drive (K2, K3)
5. Explain the v/f control of Induction motor drive (K3,K4)

Model Question paper

QP CODE:

PAGES:2

Reg.No: _____

Name: _____

**APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY
SIXTH SEMESTER B.TECH DEGREE EXAMINATION,
MONTH & YEAR
Course Code: EET 306**

Course Name: POWER ELECTRONICS

Max. Marks: 100

Duration: 3 Hours

PART A (3 x 10 = 30 Marks)

Answer all Questions. Each question carries 3 Marks

1. Explain different turn on methods of SCR.
2. Describe the reverse recovery characteristics of a power diode.
3. Draw the input and output voltage waveforms of single phase half controlled rectifier feeding RL load in continuous and discontinuous conduction mode.
4. Explain with neat sketches, the input and output voltage waveforms of 3ϕ half controlled rectifier with R load for a firing angle of 30° .
5. Compare voltage source and current source inverters.
6. Explain the terms modulation index and frequency modulation ratio related to pulse width modulation.
7. Explain time ratio control method to vary the output voltage in choppers.
8. Derive the expression for output voltage of a Buck Converter.
9. What are the advantages of electric drives?
10. Explain regenerative braking control in drives.

PART B (14 x 5 = 70 Marks)

Answer any one full question from each module. Each question carries 14 Marks

Module 1

11. a) Explain the two transistor analogy of SCR. (6)
b) Compare the switching characteristics of IGBT. (8)
12. a) Explain the structural details of MOSFET. (8)
b) Write short note on wideband gap devices. (6)

Module 2

13. a) Explain the operation of single phase full wave controlled rectifier without freewheeling diode, when feeding RL load. (10)
b) Write short notes on pulse transformer. (4)
14. a) The full-wave controlled bridge rectifier has an AC input of 220 V rms at 50 Hz and a 20 ohm load resistor. The delay angle is 40° . Determine the average current in the load, the power absorbed by the load, and the source volt-amperes. (7)
b) Draw the circuit of 3 phase fully controlled rectifier with RLE load and explain the working for $\alpha=60^\circ$ with necessary waveforms. Derive the expression for output voltage. (7)

Module 3

15. a) Explain the 120° conduction mode of a three-phase bridge inverter with output voltage waveforms, indicating the devices conducting in each state. (10)
b) Write short notes of THD. (4)
16. a) Explain sinusoidal PWM technique for varying the magnitude of output voltage in a single-phase inverter. (6)
b) Briefly explain current source inverter (8)

Module 4

17. a) Explain the working of a Buck-Boost regulator, showing relevant waveforms and derive the expression for its output voltage. (8)

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- b) Design a DC-DC Converter with 12 V input and 200 V output at upto 50 W. The ripple in the output voltage and input current should not exceed $\pm 5\%$ and $\pm 20\%$ respectively. Select suitable device and switching frequency. (6)

18. a) Describe the working of four quadrant chopper in all the four quadrants with relevant circuit diagrams. (10)

- b) Briefly explain the current limit control in dc-dc converter (4)

Module 5

19. a) Explain the working of a single phase full converter drive (8)
b) Explain the working of a four quadrant chopper drive (6)
20. a) Explain the stator voltage control for Induction motor drive (8)
b) Explain the working of v/f control of Induction motor drive (6)

Syllabus

Module 1 - 11 hrs

Introduction to Power Electronics-Scope and applications-power electronics vs signal electronics (1 hr)

Structure and principle of operation of power devices- Power diode, Power MOSFET & IGBT – switching characteristics - comparison. Basic principles of wideband gap devices- SiC, GaN (4 hrs)

SCR- Structure, Static characteristics & Switching (turn-on & turn-off) characteristics - di/dt & dv/dt protection – Turn-on methods of SCR - Two transistor analogy (5 hr)

Gate triggering circuits – Requirements of isolation and synchronization in gate drive circuits- Opto and pulse transformer based isolation (1hr)

Module 2 - 9 hrs

Controlled Rectifiers (Single Phase) – Half-wave controlled rectifier with R load– Fully controlled and half controlled bridge rectifier with R, RL and RLE loads (continuous & discontinuous conduction) – Output voltage equation- related simple problems(5 hrs)

Controlled Rectifiers (3-Phase) - 3-phase half-wave controlled rectifier with R load – Fully controlled & half-controlled bridge converter with RLE load (continuous conduction, ripple free) – Output voltage equation-Waveforms for various triggering angles (detailed mathematical analysis not required) (4 hrs)

Module 3 - 9 hrs

AC voltage controllers (ACVC) – 1-phase full-wave ACVC with R, & RL loads – Waveforms – RMS output voltage, Input power factor with R load (2 hrs)

Inverters – Voltage Source Inverters– 1-phase half-bridge & full bridge inverter with R and RL loads – THD in output voltage – 3-phase bridge inverter with R load – 120° and 180° conduction modes– Current Source Inverters-1-phase capacitor commutated CSI.(5 hrs)

Voltage control in 1-phase inverters – Pulse width modulation – Single pulse width, Multiple pulse width and Sine-triangle PWM (unipolar & bipolar modulation) – Modulation Index - Frequency modulation ratio.(2 hrs)

Module 4 - 8 hrs

DC-DC converters – Step down and Step up choppers – Single-quadrant, Two-quadrant and Four quadrant chopper – Pulse width modulation & current limit control in dc-dc converters. (4 hrs)

Switching regulators – Buck, Boost & Buck-boost –Operation with continuous conduction mode – Waveforms – Design of Power circuits (switch selection, filter inductance and capacitance) (4 hrs)

Module 5 - 11 hrs

Electric Drive: Introduction to electric drives – Block diagram – advantages of electric drives- types of load – classification of load torque (2 hrs)

DC Drives: Single phase semi converter and single phase fully controlled converter drives. Dual Converters for Speed control of DC motor-1-phase and 3-phase configurations; Simultaneous and Non-simultaneous operation. Chopper controlled DC drives- Single quadrant chopper drives- Regenerative braking control- Two quadrant chopper drives- Four quadrant chopper drives(6 hrs)

AC Drives: Three phase induction motor speed control. Stator voltage control – stator frequency control - Stator voltage and frequency control (v/f) (3 hrs)

(It is expected to emphasize the ease of independent control of field flux and armature flux in SEDC motor and relate the same with Induction motor)

Text Books

1. Muhammad H. Rashid, Power Electronics Circuits, Devices and Applications, Pearson Education
2. Daniel W. Hart, Power Electronics, Tata McGraw-Hill Education
3. P.S. Bimbhra, Power Electronics, Khanna Publishers, New Delhi

References:

1. Mohan N., T. M. Undeland and W. P. Robbins., Power Electronics, Converters,

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Applications & Design, Wiley-India

2. Fundamentals of Power Electronics, Erickson, Robert W., and Maksimovic, Dragan.
3. Krein P. T., Elements of Power Electronics, Oxford University Press, 1998.
4. L. Umanand, Power Electronics – Essentials & Applications, Wiley-India
5. Singh M. D. and K. B. Khanchandani, Power Electronics, Tata McGraw Hill, New Delhi, 2008.
6. Joseph Vithayathil, Power Electronics: Principles and Applications, McGraw-Hill College; International edition ,1995
7. Application notes on SiC and GaN, www.infineon.com. [online]
8. Evolution of wide Band-gap Semi-conductors for power devices expanding field of applications. Technical review, Vol 4, Toshiba Corporation, 2018
9. Milligan, J. W., Sheppard, S., Pribble, W., Wu, Y.-F., Muller, G., & Palmour, J. W. (2007). SiC and GaN Wide Bandgap Device Technology Overview, 2007 IEEE Radar Conference. doi:10.1109/radar.2007.374395.
10. Vedam Subramaniam “Electric drives (concepts and applications)”, Tata McGraw-Hill, 2001.
11. G. K. Dubey, Fundamentals of Electric Drives, Narosa publishers, second edition, 2010.

Course Contents and Lecture Schedule:

No.	Topic	No. of Lectures
1	Power Devices (11 hours)	
1.1	Introduction to Power Electronics: Scope and applications-power electronics vs signal electronics.	1
1.2	Structure, principle of operation, switching characteristics of Power Devices- Power Diode, Power MOSFET & IGBT – Comparison	3
1.3	Basic principles of wideband gap devices-SiC, GaN	1
1.4	SCR- Structure, Static characteristics & Switching (turn-on & turn-off) characteristics - di/dt & dv/dt protection – Turn-on methods of SCR - Two transistor analogy	5
1.5	Requirements of isolation and synchronization in gate drive circuits- Opto and pulse transformer based isolation	1
2	Single phase and three phase controlled rectifiers (9 hours)	
2.1	Half-wave controlled rectifier with R load	2
2.2	1-phase fully controlled bridge rectifier with R, RL and RLE loads (continuous & discontinuous conduction) – Output voltage equation	2
2.3	1-phase half controlled bridge rectifier with R, RL and RLE loads	1
2.4	3-phase half-wave controlled rectifier with R load	2
2.5	3-phase fully controlled & half-controlled converter with RLE load (continuous conduction, ripple free) – Output voltage equation.	2

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3	Inverters and Voltage control in single phase inverters (9 Hours)	
3.1	Applications of AC-AC converters – Single phase full-wave AC voltage controllers with R, & RL loads- Waveforms	1
3.2	RMS output voltage, Input power factor with R load	1
3.3	Voltage Source Inverters– 1-phase Half-bridge & Full bridge inverter with R and RL loads– THD in output voltage	2
3.4	3-phase bridge inverter with R load – 120° and 180° conduction modes	2
3.5	Current Source Inverters-1-phase capacitor commutated CSI.	1
3.6	Pulse Width Modulation – Single pulse width, Multiple pulse width and Sine-triangle PWM (bipolar modulation) – Modulation Index - Frequency modulation ratio.	2
4	DC-DC converters (8 Hours)	
4.1	Step down and Step up choppers – Single-quadrant chopper	2
4.2	Two-quadrant and Four-quadrant chopper – Pulse width modulation & current limit control in dc-dc converters.	2
4.3	Buck, Boost & Buck-boost – Operation with continuous conduction mode – Waveforms	3
4.4	Design of Power circuits (switch selection, filter inductance and capacitance)	1
5	Electric drives (11 Hours)	
5.1	Electric Drive: Introduction to electric drives – Block diagram – advantages of electric drives- types of load – classification of load torque	2
5.2	DC Drives: Single phase semi converter and single phase fully controlled converter drives. Dual Converters for Speed control of DC motor-1-phase and 3-phase configurations; Simultaneous and Non-simultaneous operation.	3
5.3	Chopper controlled DC drives. Single quadrant chopper drives. Regenerative braking control. Two quadrant chopper drives. Four quadrant chopper drives	3
5.4	AC Drives: Three phase induction motor speed control. Stator voltage control – stator frequency control - Stator voltage and frequency control (v/f) (3 hrs)	3